PV Remote Control User Guide

Revision History

Date	Version	Description
October 20, 2021	0.50	Initial version created.

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1 Overview

This document describes the features and usage of the second generation Photovoltaic (PV) Bluetooth LE Remote Control. The remote control showcases the features and performance of Atmosic's ATM3 energy harvesting Bluetooth LE IC. The remote control is designed around the ATM3, Photovoltaic (PV) cells to harvest energy from ambient light, as well as a coin cell battery and supercapacitor for energy storage.

The ATM3 combines ultra low power Bluetooth LE with an integrated Power Management Unit (PMU) that enables direct connections to energy harvesting elements like PV cells. The direct connection to the ATM3 PMU has multiple benefits, reducing the overall cost of adding energy harvesting and improving the efficiency of harvested energy usage. In addition, the low power consumption of the ATM3 means a smaller PV cell is needed for the remote, further reducing both the cost and area required for the PV cell.

The PV Remote Control is designed to operate at indoor light levels primarily from harvested light with excess harvested energy stored in the super capacitor. The CR2032 coin cell battery is only utilized when sufficient harvested and stored energy is not available, such as during extended periods of operation in a dark environment. Efficiently using harvested energy as much as possible enables the battery life of the remote control to be greatly extended.

The PV Remote Control is provided for demonstration and evaluation purposes only and is not a commercial product for sale. A list of feature limitations and future enhancements is provided at the end of this document.

NOTE: This document does not cover the first generation PV Remote Control. If you have an earlier version, please contact your local Atmosic representative for additional information and support.

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2 PV Remote Evaluation Kit

2.1 PV Remote Control

The front of the PV Remote Control is shown in Figure 1. The PV panel is a 10.9 cm^2 amorphous Silicon PV cell from Panasonic. The CR2032 battery is accessible under a cover on the back of the remote as depicted in Figure 2. Some of the other remote features:

- Compliance with Google Voice over Bluetooth LE specification V0.4e
- 8 KHz/16-bit IMA ADPCM voice Encoder
- 16 keys configured in a 4x4 key matrix

NOTE: The PV Remote Control is designed to be used only with the ADT-3 device.

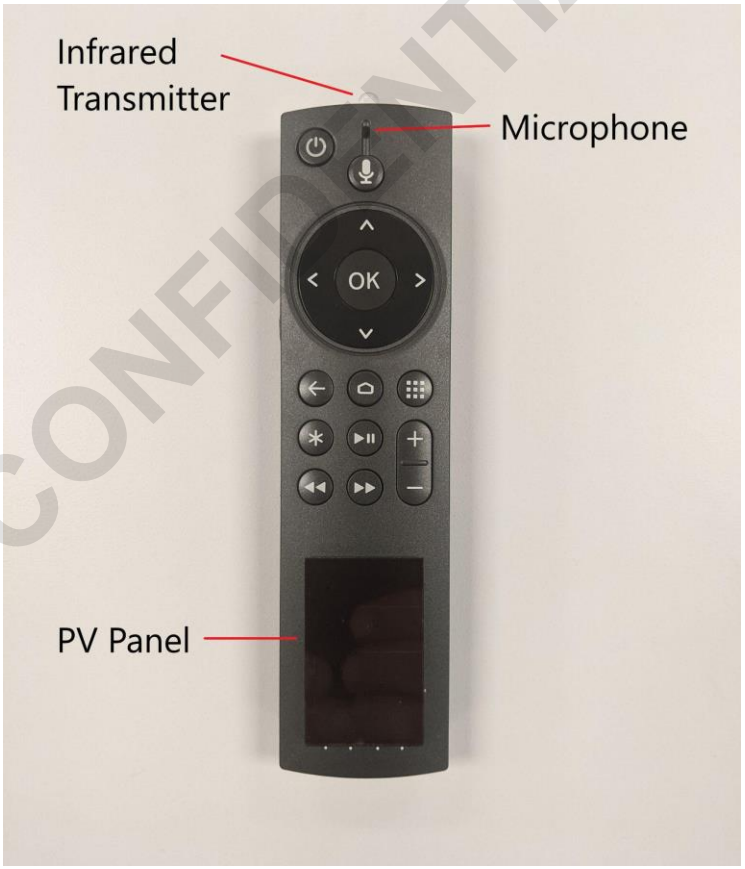


Figure 1 - Front of PV Remote Control

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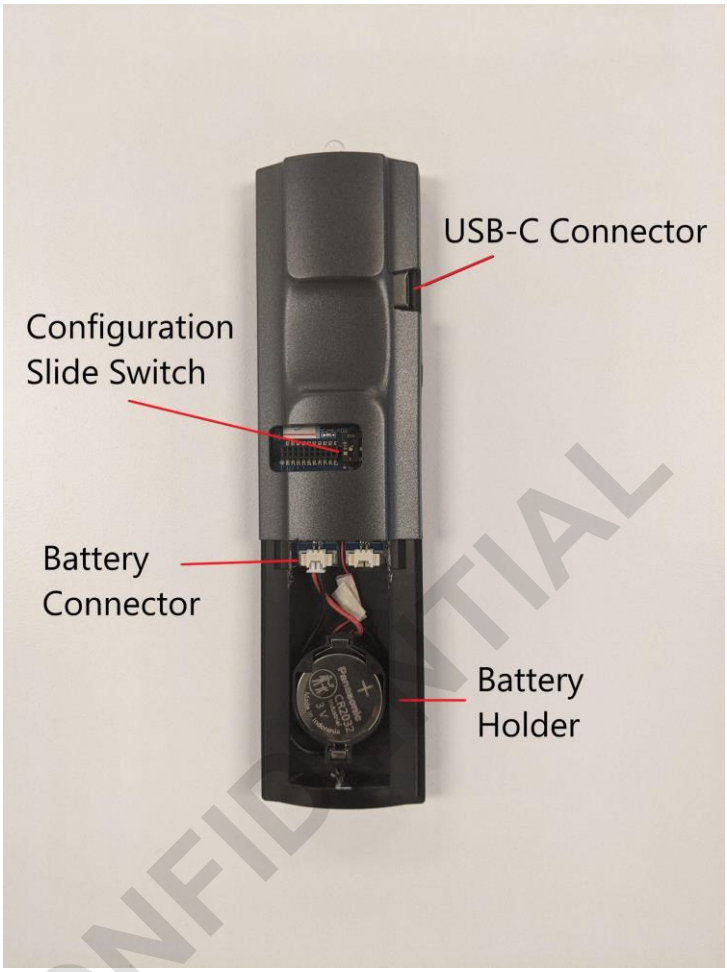


Figure 2 - Back of Remote with Battery Access

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The configuration slide switch configures the remote for different battery and energy storage configurations. The supported configurations are covered in section 3.5 Slide Switch Configurations..

The major electrical components of the remote are depicted in the block diagram shown in Figure 3.

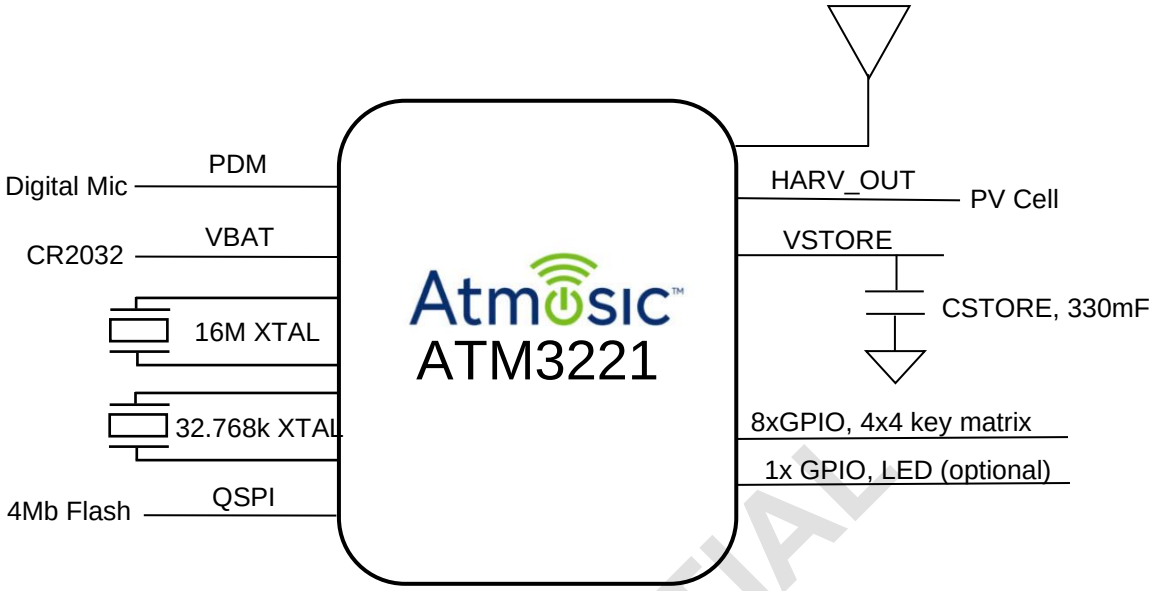


Figure 3 - PV Remote Block Diagram

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3 Remote Control Operation

3.1 General Operation

The state machine for the remote control is detailed in Figure 4. Upon power-up the remote will automatically enter the pairing state if it has not already been paired with a device. If pairing information is present it will attempt to reconnect and enter a hibernation state if it cannot connect in 30 seconds. A keypress on the remote is required to exit the hibernation state and restart the reconnection or pairing process.

Once connected the remote has a default connection interval of 10 ms with a slave latency setting of 99.

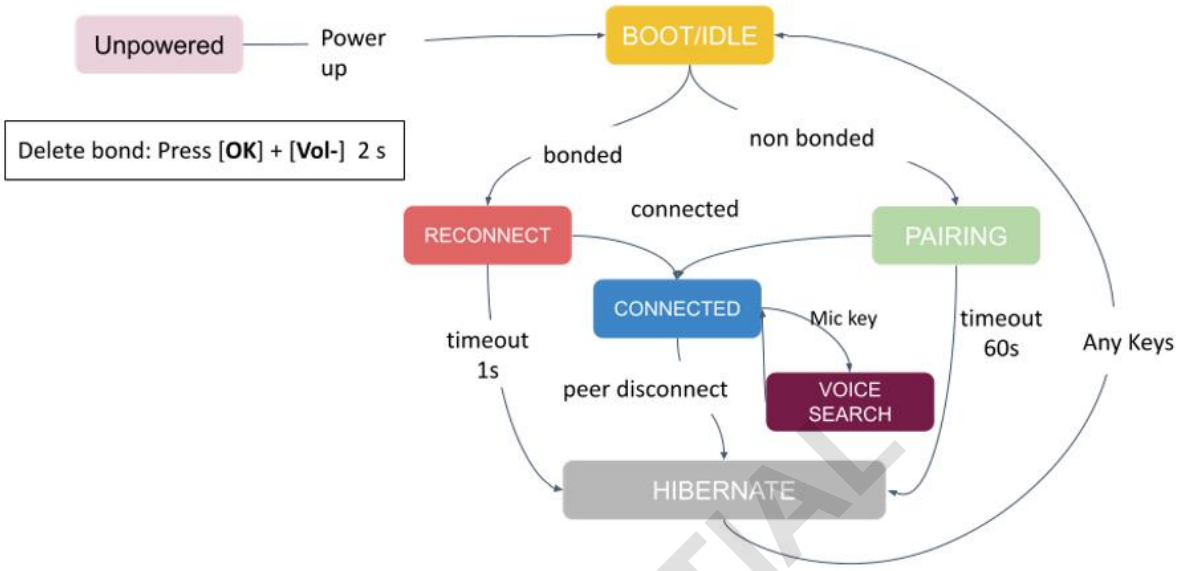


Figure 4 - Remote Control State Diagram

3.2 Pairing the Remote

To pair the remote it is necessary to use the connect a Bluetooth device option in the settings menu of the Android ADT. The existing remote will be needed to select this option and pair with the "Atmosic remote xxx" device. If the Atmosic remote device is not found it may either already be paired with another device or in the hibernate state. Press a key on the remote to exit from the hibernation state and/or follow the instructions to unpair the remote found in the next section.

After startup, if pairing is not completed in 60 seconds the remote will enter the hibernate state until a key is pressed to return it to the pairing state.

3.3 Unpairing the Remote

The remote can be unpaired by pressing the "OK" and "-" keys simultaneously for two seconds.

3.4 Remote LED

The remote has a red LED visible through the microphone hole at the very top of the remote. The LED provides information about the current operating status of the remote. A list of valid LED behaviors is provided in Table 1.

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Operating State	LED Behavior
Pairing	Blink at 250 ms interval
Reconnecting	Blink at 100 ms interval
Remote key press (connected)	On for 100 ms

Table 1 - LED Operation

3.5 Slide Switch Configurations

A set of slide switches is accessible through the opening on the back side remote as shown in Figure 1. The slide switches are used to change operating modes for the battery and supercapacitor connections. A list of acceptable configurations is provided in Table 2.

Battery type	Slide Switch-1	Slide Switch-2	Slide Switch-3
PV cell with CR2032 and Supercap	OFF	OFF	ON
PV cell with supercapacitor disconnected	OFF	OFF	OFF

Table 2 - Slide Switch Configurations

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4 Remote Power Consumption

The power consumption of the Atmosic remote control solution in the three most frequent application states is listed in Table 3. These measurements were taken at the battery connection during actual operation with the PV cell and supercapacitor disconnected as energy sources.

Operating State	Current Consumption
Connected to TV, Idle	5.55 μA
Single key press	270 μA
Voice Input	1.15 mA

Table 3 - Remote Control Power Consumption

It is possible to duplicate these measurements by covering up the PV cell and disconnecting the supercapacitor as an energy storage device. Refer to Table 2 for the correct slide switch configuration to disconnect the supercapacitor as an energy source.

The current consumption is measured at the battery connection to the remote circuit board The connector is indicated by the **red** circle in Figure 5. An inline instrument such as a Joulescope or a separate precision power source can be connected to make the measurements.



Figure 5 -Battery Connection for Current Consumption Measurement

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NOTE: The supercapacitor must be disconnected to give an accurate measurement of the remote power consumption. If the supercapacitor is connected and contains any energy it will be used and skew the power consumption measurement results.

Atmosic has developed various analysis tools to help customers estimate the daily energy consumption of the remote control based on a model of daily usage as well as tools to analyze the performance of the remote based on different PV cell types and sizes; battery technology and size; supercapacitor size; and ambient light levels. For more information contact your Atmosic support or your local Atmosic sales representative.

5 Loading New Firmware

A new firmware image is loaded into the remote control with the Atmosic ISP GUI Tool running on a PC connected to the remote via the provided interface board and cable.

The instructions for loading new firmware are as follows:

1. Remove the remote's battery from the battery holder and connect the remote to the provided interface board using the 20 pin cable as shown in Figure 6.
2. Connect the interface board to the PC using the provided USB cable. Plug the USB cable

into the USB1 port of the interface board as shown in Figure 6.

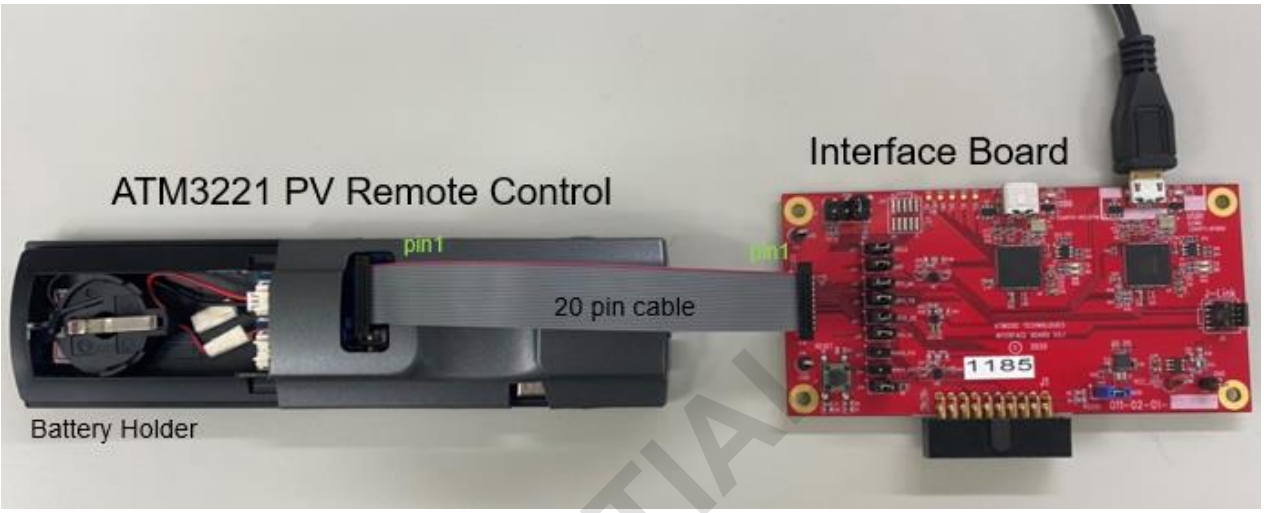
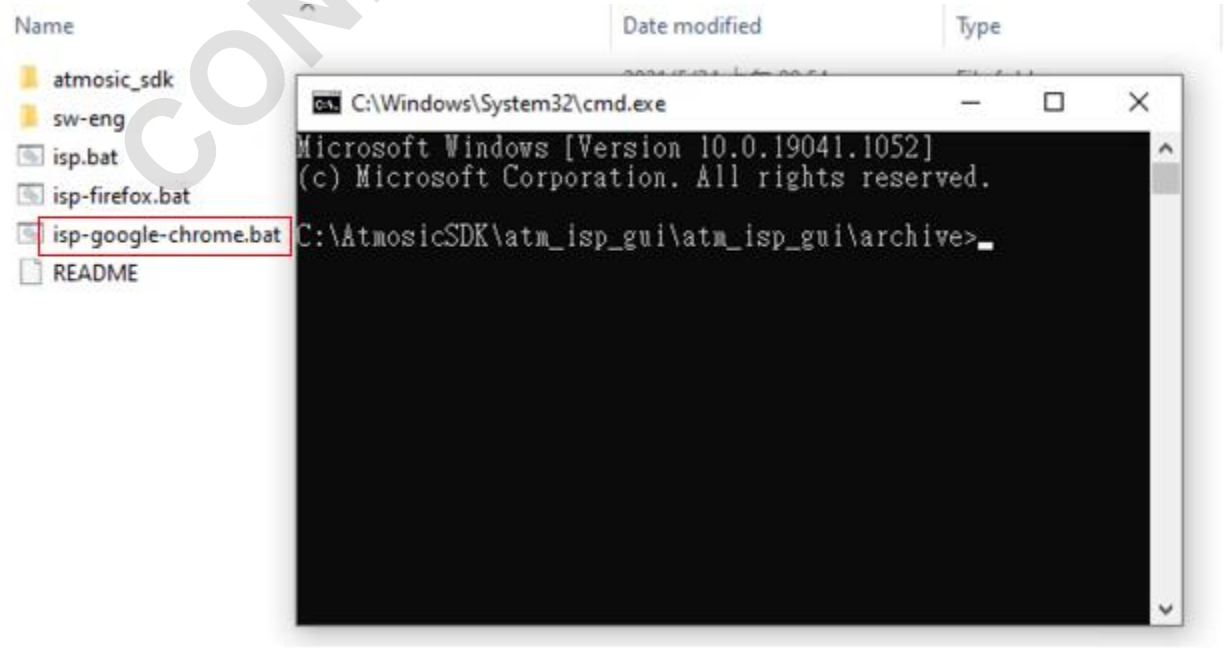


Figure 6 - Hardware Setup for Firmware Update

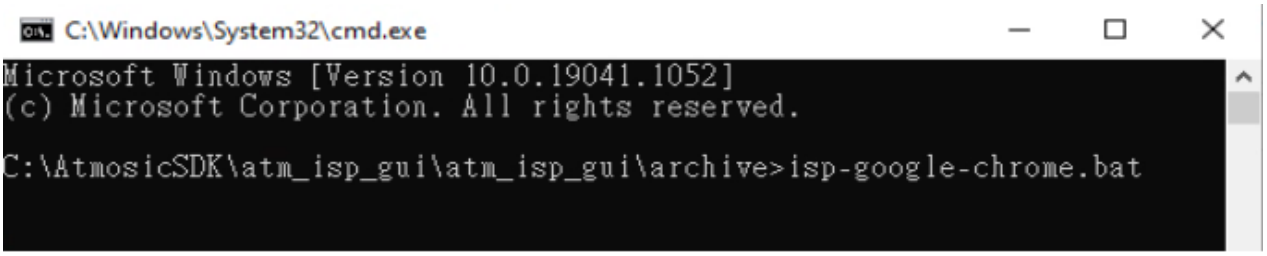
3. The Atmosic ISP GUI Tool is provided with the Atmosic Software Development Kit and can be downloaded from the support section of the Atmosic website. If you require assistance please contact Atmosic support (support@atmosic.com).

4. Unzip the atm_isp_gui.7z file, open a command window, and navigate to the location of the isp-google-chrome.bat file.



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5. Make sure that the remote is not in hibernate mode (enter pairing mode on the remote if necessary) and execute the isp-google-chrome.bat command in the command window.



6. If a security alert appears, press the *Allow access* button to continue

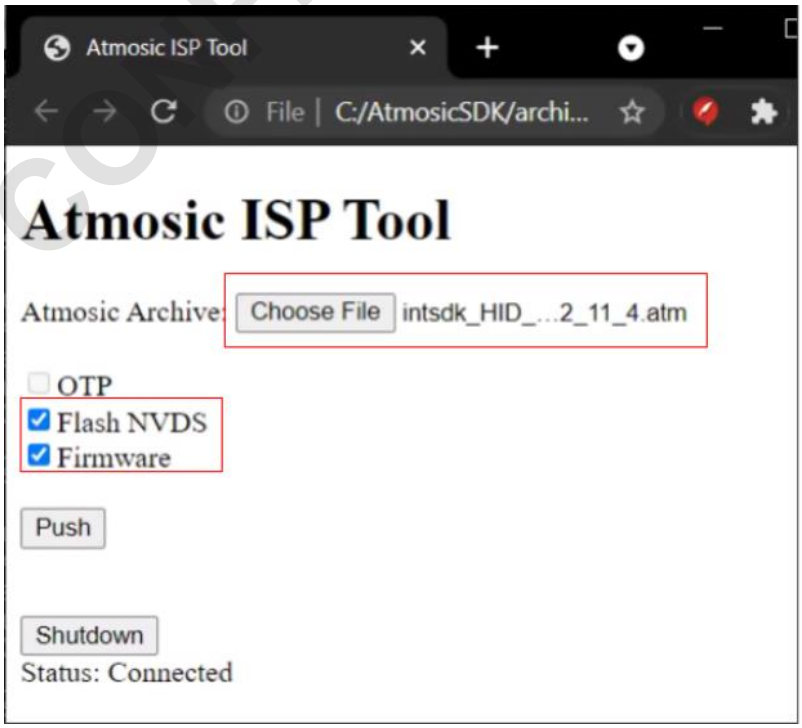


7. The ISP Tool will begin execution in a browser window. Wait until the Status indicator shows *Connected.* See the red outlined box in the figure below.

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8. Select *Choose File* to select the new .atm firmware image. Once selected the *Flash NVDS* and *Firmware* boxes will be checked as shown below.

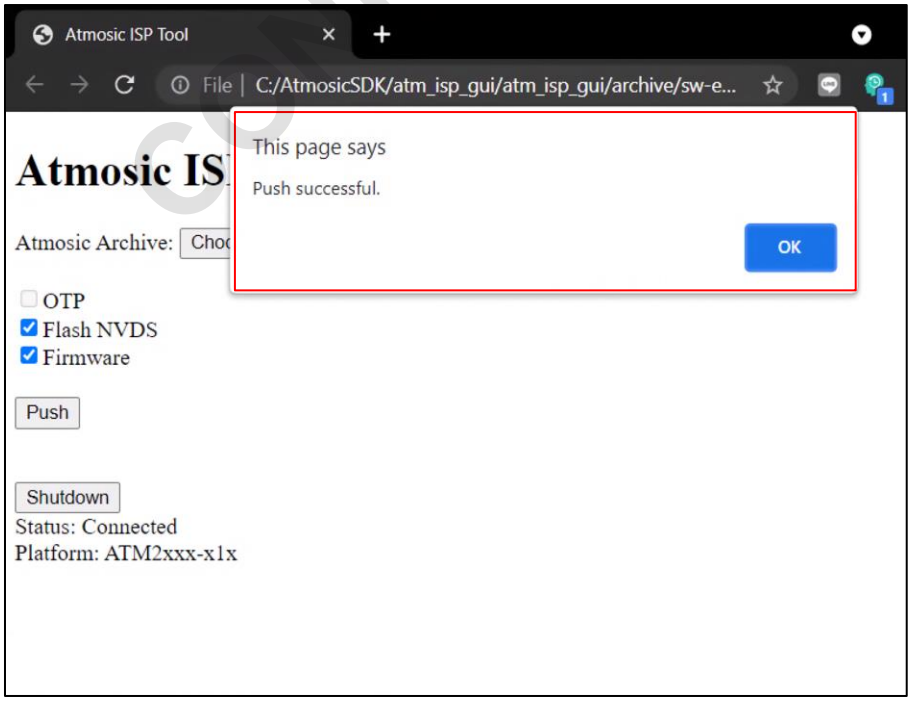


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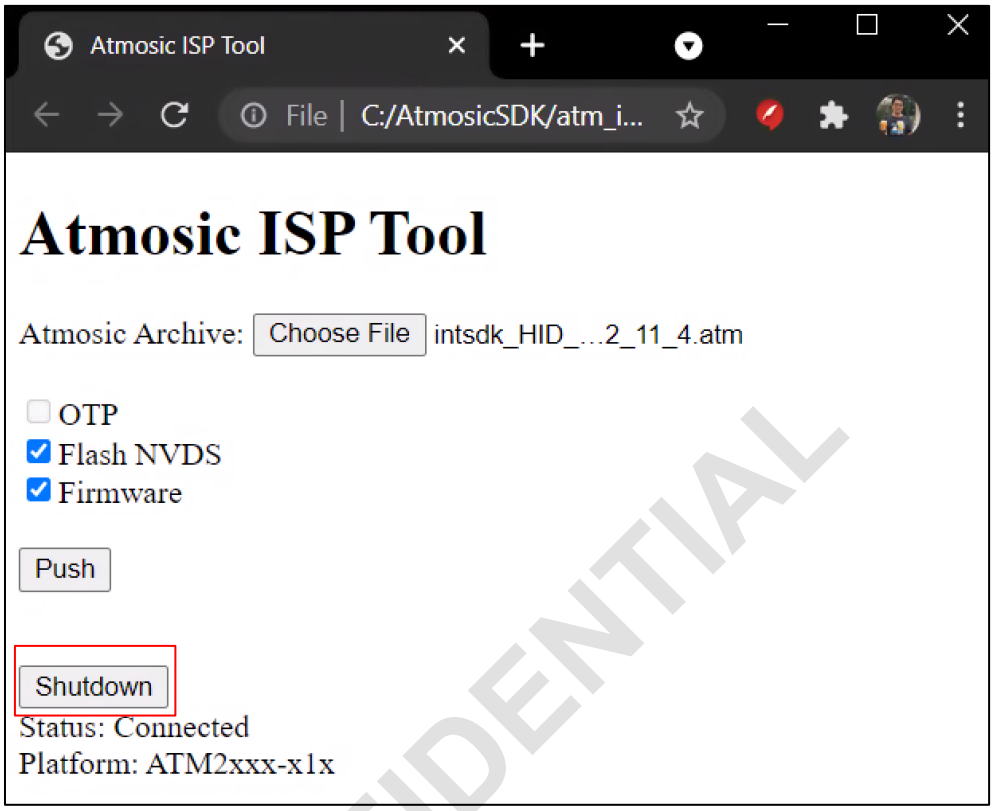
9. Select *Push* to begin the update and a pop-up window will appear showing the update progress.

```
Info : SWD DPIDR 0x0bb11477
Info : Sydney.cpu: hardware has 4 breakpoints, 2 watchpoints
target halted due to debug-request, current mode: Thread
xPSR: 0xc1000000 pc: 0x00005530 msp: 0x20014000
target halted due to debug-request, current mode: Thread
xPSR: 0xc1000000 pc: 0x00005530 msp: 0x20014000
Macronix 4Mb flash (524288 bytes)
53 bytes written at address 0x10078000
downloaded 53 bytes in 0.016169s (3.201 KiB/s)
INFO:     'sydney_load_nvds flash_nvds_1623.bin 491520 32768' executed
INFO:     Client connected: 127.0.0.1
INFO:     Responding.
INFO:     Platforms: ATM2xxx-x1x
INFO:     Client connected: 127.0.0.1
INFO:     Responding.
INFO:     Running OpenOCD SVC command sydney_load_flash image
Polling target Sydney.cpu failed, trying to reexamine
Info : SWD DPIDR 0x0bb11477
Info : Sydney.cpu: hardware has 4 breakpoints, 2 watchpoints
target halted due to debug-request, current mode: Thread
xPSR: 0xc1000000 pc: 0x00005530 msp: 0x20014000
target halted due to debug-request, current mode: Thread
xPSR: 0xc1000000 pc: 0x00005530 msp: 0x20014000
Macronix 4Mb flash (524288 bytes)
```

10. Once completed, a pop-up window will appear to indicate the update was successful.



11. Select *Shutdown* to exit the tool



12. Disconnect the 20 pin cable from the remote and re-insert the battery to start execution of the new firmware image.

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6 Remote Control Limitations

The PV remote control as provided does not support the following features:

1. Rechargeable batteries - Atmosic continues to add new battery options to the PV remote control application. Alternatives to the standard CR2032 coin cell require a configuration change and new firmware.
2. Infrared (IR) support - by default, the PV remote firmware does not transmit any data via IR. Support for IR requires a new firmware image.

If support for these features is desired, contact your local Atmosic representative for assistance.



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